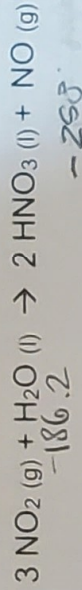


3. Use standard enthalpies of formation to calculate the enthalpy change for the following reaction. (4 marks)



-186.2
-258

Products

Reactants

$$\Delta H = (3\Delta H_{\text{HNO}_3} + \Delta H_{\text{H}_2\text{O}}) + (2\Delta H_{\text{HNO}_3} + \Delta H_{\text{NO}})$$

$$= (3(-186.2) + (-285.8)) + (2(-174.1) + 90.2)$$

$$\Delta H = -(-186.2) + (-258.4 \text{ kJ})$$

$$\Delta H_{\text{f,rxn}} = -71.8 \text{ kJ}$$

COMMUNICATION (8 marks)

1. A 62.16 g sample of molten lead solidifies, exchanging 98.1 kJ of energy with the surroundings.

- a) What is the molar enthalpy of melting, ΔH_{melt} , for lead? (2 marks)

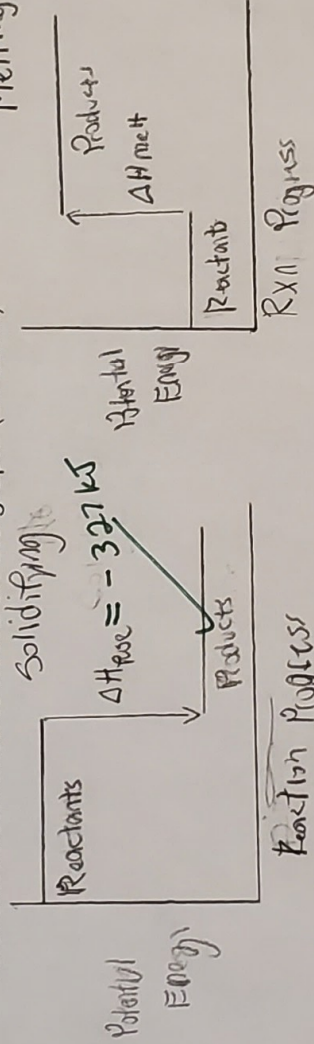
$$\Delta H_{\text{h}} = \frac{\Delta H}{n} \quad n = \frac{62.16 \text{ g}}{207.2 \text{ g/mol}} \quad \Delta H_{\text{fuse}} = -327 \text{ kJ/mol}$$

$$\Delta H_{\text{melt}} = +327 \text{ kJ/mol}$$

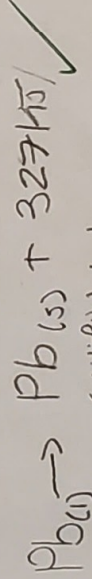
$$\Delta H_{\text{h}} = -327 \text{ kJ/mol}$$

- b) Classify the solidification of molten lead as endothermic or exothermic. Exothermic (1 mark)

- c) Sketch a potential energy diagram for the reaction. Ensure that all components are labelled and all relevant information is noted on the graph. (2 marks)



- d) Write a thermochemical equation for this reaction. (1 mark)



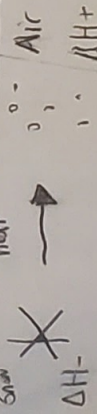
Solidifying Lead

2. On a calm winter day, it begins to snow and the air temperature increases. Explain the enthalpy change occurring. (2 marks)

The snow when freezing expels energy as it is an exothermic reaction.

$\Delta H_{\text{snow}} = -q$. The air (surroundings) gains the released heat of the snow.

$$\Delta H_{\text{snow}} = +q \quad \text{Thus } q_{\text{snow}} + q_{\text{air}} = 0, \text{ so energy is conserved.}$$



Higher potential

Snow is at a lower thermal energy while air is at a higher thermal energy.

THINKING (15 marks)

Great job!

- In a coffee cup calorimeter, a student mixes 1.60 g of ammonium nitrate, $\text{NH}_4\text{NO}_3(\text{s})$, with 75.0 g of water at an initial temperature of 25.0°C . After dissolution of the salt, the final temperature of the calorimeter content is 23.7°C . Calculate the **molar enthalpy** for the dissolution of ammonium nitrate. Assume the density and specific heat capacity of the resulting solution is similar to that of liquid water, $4.184 \text{ J/g}\cdot^\circ\text{C}$. (5 marks)

USE GRASP!

G $T_F = 23.7^\circ\text{C}$ $T_I = 25^\circ\text{C}$ $m_{\text{water}} = 75\text{g}$ $m_{\text{soln}} = 100\text{g}$ $C_{\text{water}} = 4.184 \text{ J/g}\cdot^\circ\text{C}$

R $\Delta H_n = ?$

A $q_{\text{water}} = cm\Delta T$ $\Delta H = -q_{\text{water}}$ $\Delta H_n = \frac{\Delta H}{n_{\text{NH}_4\text{NO}_3}}$ $n = \frac{m}{M}$

S $q_{\text{water}} = (4.184 \text{ J/g}\cdot^\circ\text{C})(76.6\text{g})(-1.3^\circ\text{C})$ $n = \frac{1.60\text{g}}{140.0\text{g/mol} + 4(1.01\text{g/mol}) + 3(16.0\text{g/mol})} = 0.0119\text{mol}$

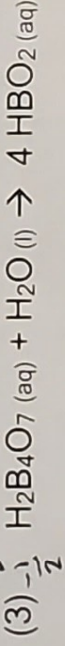
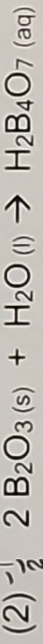
$q_{\text{water}} = -416.24\text{J}$

$\Delta H = 416.24\text{J}$

$\Delta H_n = \frac{\Delta H}{n} = \frac{416.24\text{J}}{0.0119\text{mol}} = 20.9\text{kJ/mol}$

Final molar enthalpy for NH_4NO_3

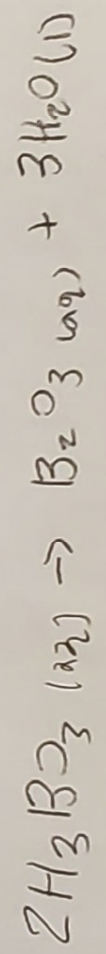
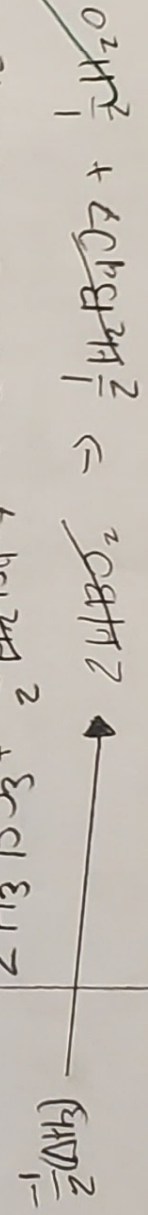
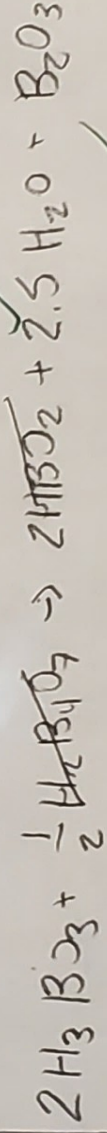
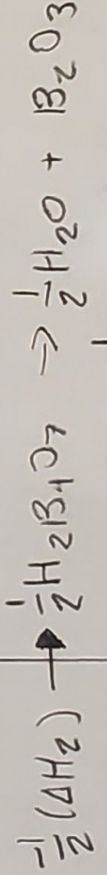
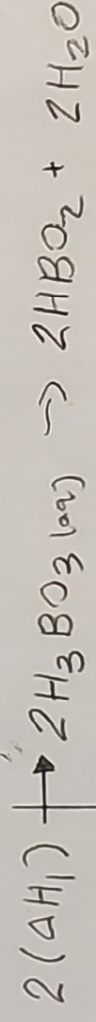
- Use the following thermochemical equations to calculate the enthalpy change for the following reaction. Show all your work! (6 marks)



$\Delta H_1 = -0.02 \text{ kJ}$

$\Delta H_2 = -17.5 \text{ kJ}$

$\Delta H_3 = -11.3 \text{ kJ}$



$\Delta H = 2(-0.02\text{kJ}) + \frac{1}{2}(-17.5\text{kJ}) + \frac{1}{2}(-11.3\text{kJ})$

$\Delta H = 14.36 \text{ kJ}$